

HelpDeskOps Intelligence

Helpdesk Operations Analytics Report

*Prepared using Python • SQL • Power BI
Operational Analytics Case Study (2016–2023)*

1. Executive Summary

The HelpDeskOps Intelligence dashboard analyzes helpdesk operations from 2016–2023 to evaluate ticket demand, backlog, SLA performance, and agent productivity. The analysis covers approximately 42,974 tickets and highlights operational bottlenecks that directly impact service quality. Results show that Medium and High priority tickets contribute most of the backlog, overall SLA compliance remains low (24.57%), and ticket assignment is uneven across agents. These findings suggest opportunities to improve workload balancing, backlog reduction, and SLA adherence.

2. Project Overview

Objective: Build an operational analytics dashboard that enables support managers to monitor ticket flow, identify bottlenecks, measure service performance, and make data-driven decisions.

3. Key Performance Indicators

Metric	Value
Total Tickets	42,974
Resolved Tickets	42K
Backlog Tickets	853
Average Ticket Age	2.54K Days
SLA Compliance	24.57%
SLA Breached	7K
Total Agents	375
Average Tickets / Agent	114.60

4. Dashboard Findings

Support Overview

Ticket volume peaked in 2016 and 2022 before declining in 2023. Medium and High priority tickets account for most backlog, while several issue types exhibit extremely high ticket ages, indicating long-standing unresolved work.

Workload & Backlog

The active queue contains significant Waiting and Backlog tickets. Medium-priority work dominates outstanding requests, suggesting that operational attention is concentrated on urgent tickets while medium-priority issues accumulate.

SLA & Resolution

Only 24.57% of resolved tickets met SLA requirements. Highest and Blocker priorities perform comparatively better, whereas Ticket and Service issue types account for most SLA breaches.

Agent Performance

Ticket distribution is highly uneven. One agent manages substantially more tickets than peers, indicating workload imbalance that may contribute to slower resolutions and inconsistent service quality.

5. Business Insights

- High ticket demand and backlog create sustained operational pressure.
- Long ticket age is strongly associated with poor SLA performance.
- Medium-priority tickets require proactive backlog management.
- Uneven workload distribution suggests opportunities for smarter ticket routing and resource planning.

6. Recommendations

- Introduce backlog aging buckets (0–7, 8–30, 31–90, 90+ days).
- Prioritize aging Medium and High priority tickets.
- Monitor SLA performance by priority, issue type, and agent.
- Redistribute workload across agents to improve operational efficiency.
- Use true resolution timestamps in future versions to replace ticket-age proxies.

7. Conclusion

HelpDeskOps Intelligence demonstrates how operational data can be transformed into actionable business insights. The dashboard enables managers to monitor ticket demand, backlog, SLA compliance, and workforce productivity, supporting better operational decisions. The project showcases practical skills in data cleaning, SQL analytics, KPI design, Power BI visualization, and business storytelling.